

PRICING AI AS A SERVICE

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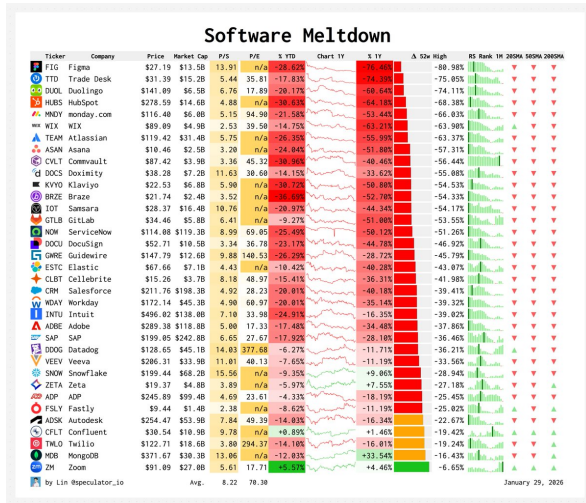
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SOFTWARE AS A SERVICE (SAAS)

- SaaS monetized access to software through seats and subscriptions

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WHY SAAS IS BEING REPRICED

- AI agents are becoming an execution layer inside business workflows.
- Examples:
 - resolve customer service requests (Zendesk, Intercom)
 - draft and follow up on sales workflows (Salesforce, HubSpot)
 - process procurement requests (Coupa)

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What buyer really cares about is no longer just access to the technology itself — it is whether the AI agent can actually finish the task.

PRICING FOR AI IS MOVING BEYOND USAGES!

Understanding outcome-based pricing results-driven framework

Learn how outcome-based pricing ties AI investment to measurable results, and evaluate value based on what agents achieve, not just what they cost.

AI pricing: AI is Driving A Shift Towards Outcome-Based Pricing

 FlyCode Inc.
2,129 followers

February 19, 2025

In recent months, we've witnessed a paradigm shift in how businesses price their products and services. The traditional subscription or usage-based pricing models are giving way to something more dynamic, more results-oriented, and more aligned with customer success: outcome-based pricing. At the heart of this transformation is artificial intelligence (AI), which enables companies to not only deliver measurable outcomes but also quantify and monetize their value effectively.

Why Outcome-Based Pricing?

Outcome-based pricing flips the script on how value is measured and delivered. Instead of customers paying for access, licenses, or hours of use, they're paying for results. This approach has gained momentum as businesses demand greater ROI and alignment between their spending and the tangible outcomes they achieve.

Metronome Webinar Recap Outcome-based pricing

What is Outcome-Based Pricing, and How Can You Use It?

MARCH 12, 2025 • 61 MIN READ

 Gregorian Kemp
Content Marketing

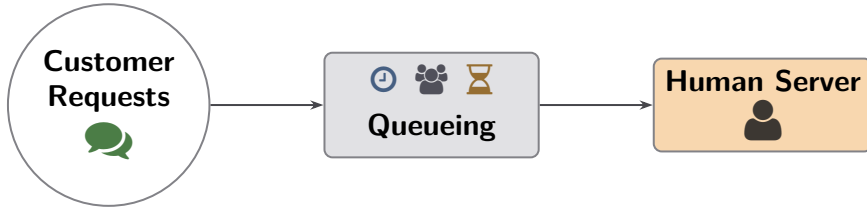
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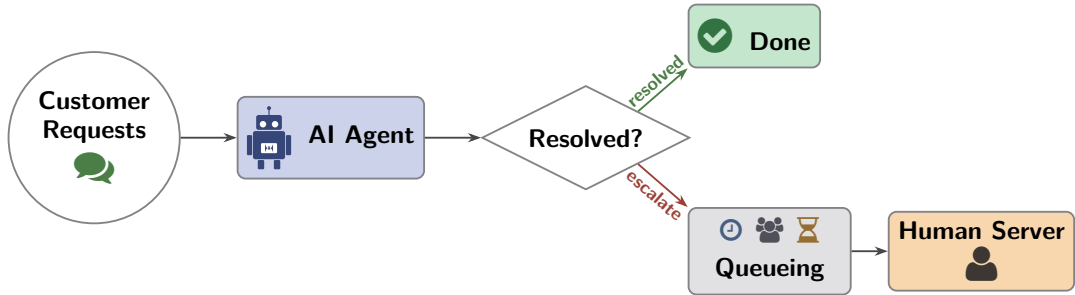
Outcome-Based Pricing Is Eating the Software Industry

YouTube · Mostly Growth
2 days ago

EXAMPLE: CUSTOMER SERVICE WITHOUT AI AGENTS



EXAMPLE: AFTER DEPLOYING AI AGENTS

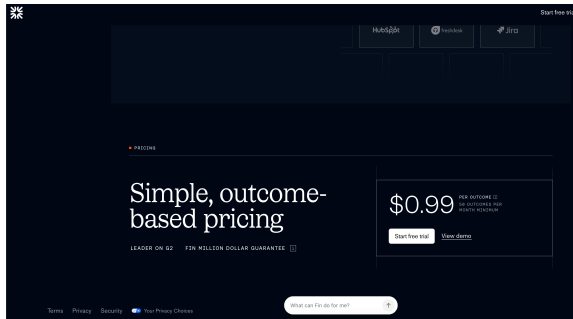


A FUNNY FACT

Anthropic's customer support service is outsourced to Fin AI...



Claude



Fin AI: pay only for success

PAY-ONLY-ON-SUCCESS PRICING: APPEAL AND TENSION

Why it is appealing

- Aligns the bill with realized business value $\Rightarrow$ easier adoption.

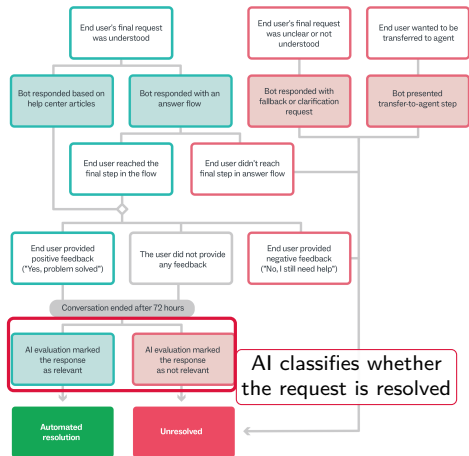
PAY-ONLY-ON-SUCCESS PRICING: APPEAL AND TENSION

Why it is appealing

- Aligns the bill with realized business value $\Rightarrow$ easier adoption.

Why it is hard

- “Success” is **reported by the same party that gets paid for it**.
- The seller can tune *when a conversation counts as resolved*, classify handoffs, or calibrate the AI’s display.



Source: Zendesk Help Center, “About automated resolutions for AI agents.”

BUYERS' CONCERNS ABOUT MISREPORTING ARE REAL

“outcome-based pricing sounds great until you realize everyone defines “outcome” differently lol, usage is at least objective even if customers complain about it less.”

Sources: Reddit r/AI_Agents, “Outcome-based pricing”; company pricing pages.

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Usage-based pricing

- Salesforce Agentforce
- Tidio Lyro

Pay-for-success pricing

- Intercom Fin
- Zendesk AI Agents
- Sierra / Gorgias

Sources: Reddit r/AI_Agents, “Outcome-based pricing”; company pricing pages.

RESEARCH QUESTION

What are the revenue and incentive implications of pricing AI based on outcomes?

Key challenge: it is not always obvious whether the AI actually accomplished the task.

RESEARCH QUESTION

What are the revenue and incentive implications of pricing AI based on outcomes?

Two concrete questions:

- When is pay-only-on-success better than usage-based pricing?
- What is the optimal pricing and outcome-reporting mechanism?

MODEL PRIMITIVES

- Seller offers AI models/agents with capability $a \in \mathcal{A} \subseteq \mathbb{R}_+$
 - with increasing and convex cost $c(a)$

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- Seller offers AI models/agents with capability $a \in \mathcal{A} \subseteq \mathbb{R}_+$
 - with increasing and convex cost $c(a)$
- Buyer has private type $t \in \mathcal{T} = \{1, \dots, n\}$, with prior $\pi_t > 0$.
- Binary true outcome:

$$\omega \in \{\omega_0, \omega_1\} \text{ (failure/success)}$$

- Type t buyer when assigned an AI model with capability a :

$$v_t(a) = \text{valuation}, \quad \gamma_t(a) = \Pr(\omega = \omega_1 \mid t, a).$$

- In a *static* setting,

$$v_t(a) = \gamma_t(a) \cdot \underbrace{\tilde{v}_t(\omega_1)}_{\text{success value}} + (1 - \gamma_t(a)) \cdot \underbrace{\tilde{v}_t(\omega_0)}_{\text{failure value}}$$

CONTRACT AND OUTCOME-REPORTING MECHANISM

CONTRACT

For each report t , the seller assigns:

$$(a_t, p_t, q_t).$$

- a_t : capability.
- p_t : bill after displayed success.
- q_t : bill after displayed failure.

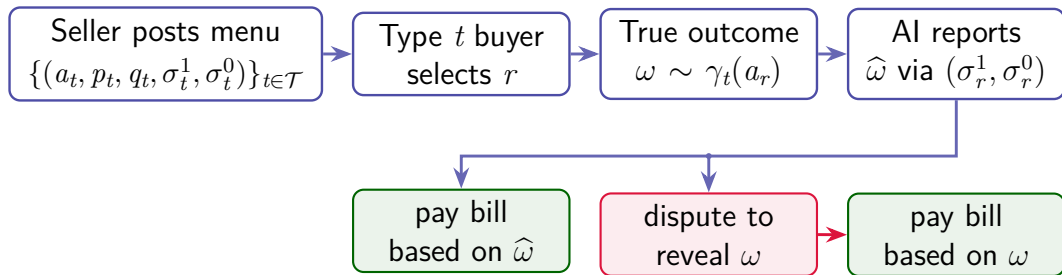
OUTCOME-REPORTING MECHANISM

For each report t , the AI's outcome-reporting rule uses:

$$(\sigma_t^1, \sigma_t^0).$$

$$\begin{aligned}\sigma_t^1 &= \Pr(\hat{\omega} = \omega_1 \mid \omega = \omega_1, t), \\ \sigma_t^0 &= \Pr(\hat{\omega} = \omega_1 \mid \omega = \omega_0, t).\end{aligned}$$

TIMING



verification costs κ_B, κ_S

- κ_B : buyer-side verification cost
- κ_S : seller-side verification cost

SELLER'S PROBLEM

$$\max_{\{(a_t, p_t, q_t, \sigma_t^1, \sigma_t^0)\}_{t \in \mathcal{T}}} \sum_t \pi_t \left[\mathbb{E}\{\text{bill}_t\} - \kappa_S \Pr\{\text{dispute}_t\} - c(a_t) \right].$$

- [SR] : after observing $\hat{\omega}$, buyer optimally pays or disputes,
- [IC] : truthfully report private type t beats any misreport,
- [IR] : expected utility is nonnegative.

WHY THE PROBLEM IS HARD

JOINT DESIGN

A joint contract design + information design problem. The seller chooses:

$$\underbrace{(a_t, p_t, q_t)}_{\text{contract}} \quad \underbrace{(\sigma_t^1, \sigma_t^0)}_{\text{signaling}}.$$

BEYOND 1-D MECHANISM DESIGN

Type t is 1-dimensional, but maps to two payoff-relevant functions:

$$t \longmapsto \left(\underbrace{v_t(\cdot)}_{\text{valuation}}, \quad \underbrace{\gamma_t(\cdot)}_{\text{success probability}} \right).$$

RELATED WORKS (CONDENSED)

- **Non-linear pricing (screening)**

- Mussa and Rosen (1978); Maskin and Riley (1984); Myerson (1981), Haghpahanah and Siegel (2025)...

- **Information design**

- Kamenica and Gentzkow (2011); Bergemann and Morris (2016)...

- **Economics of AI**

- Dütting et al. (2024), Bergemann, Bonatti and Smolin (2025)....

- **Mechanism design with post-allocation signals**

- Welfare / efficient implementation: Mezzetti (2004), Bergemann et al. (2025)
- **revenue maximization:** Riordan and Sappington (1988), Alaei et al. (2026).
- **this work: outcome reporting + contract design with post-allocation signals**

WARMUP: TWO SIMPLE PRICING FORMATS

USAGE-BASED PRICING (UBP)

UBP : $p_t = q_t$.

- Same bill after displayed success/failure.
- This collapses to standard screening.

PAY-FOR-SUCCESS PRICING (PFS)

PFS : $q_t = 0$.

- Pay only on displayed success.
- Outcome reporting and dispute incentives matter.

OUTCOME REPORTING IN SIMPLE PRICING FORMATS

REMARK (OUTCOME REPORTING)

- **UBP:** $p_t = q_t$. The displayed outcome does not change the bill, so outcome misreporting is unnecessary.

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REMARK (OUTCOME REPORTING)

- **UBP:** $p_t = q_t$. The displayed outcome does not change the bill, so outcome misreporting is unnecessary.
- **PFS:** $q_t = 0$. Displayed success triggers payment, so truthful outcome reporting is generally **NOT** optimal.

COMPARISON: PFS vs. UBP

THEOREM (COMPARISON BETWEEN PFS AND UBP)

If the buyer-side dispute cost κ_B is high, then

PFS + misreporting $\iff$ LL second best $(p, q \geq 0)$ + truthful revealing.

Thus, $R_{\text{PFS}}^* \geq R_{\text{UBP}}^*$.

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Intuition.

- High buyer-side dispute cost: displayed successes go largely unchallenged.
- False-success reports after true failure create an expected failure-state charge.
- This gives PFS more billing flexibility than UBP.

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- If the buyer-side dispute cost κ_B is small (*informal*): $R_{\text{PFS}}^* \geq R_{\text{UBP}}^*$ when $\frac{v_t(a)}{\gamma_t(a)}$ aligns with $\gamma_t(a)$; otherwise, $R_{\text{UBP}}^* \geq R_{\text{PFS}}^*$.

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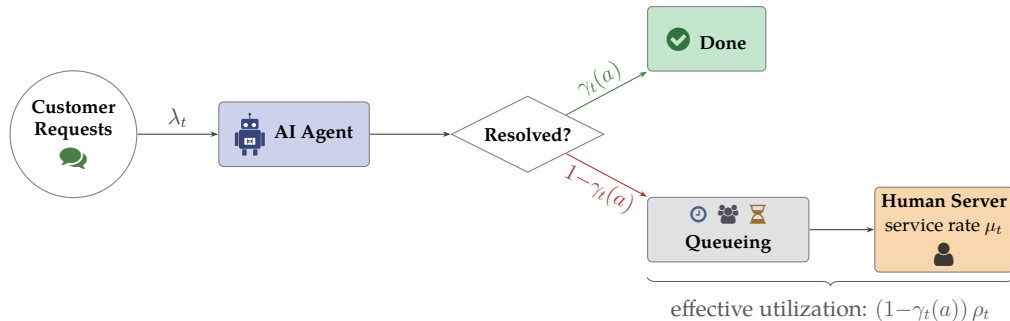
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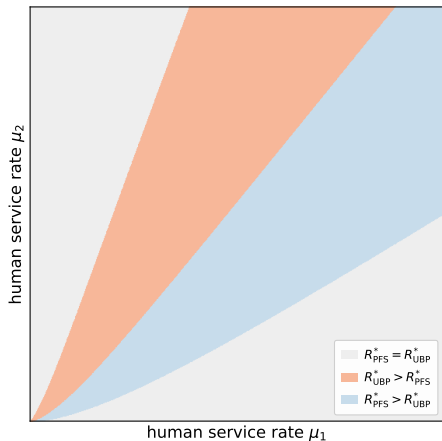
- Alignment means easier AI tasks also have higher success value for the buyer.

APPLICATION: PRICING AI IN CUSTOMER SERVICE



- Type t has arrival rate λ_t , human service rate μ_t , and baseline utilization $\rho_t = \lambda_t / \mu_t$.
- AI reduces the steady-state load sent to humans: $\rho_t \longrightarrow (1 - \gamma_t(a)) \rho_t$.
- Buyer value $v_t(a)$ is the *resulting reduction in steady-state congestion cost*.

UBP vs. PFS IN THE QUEUEING APPLICATION

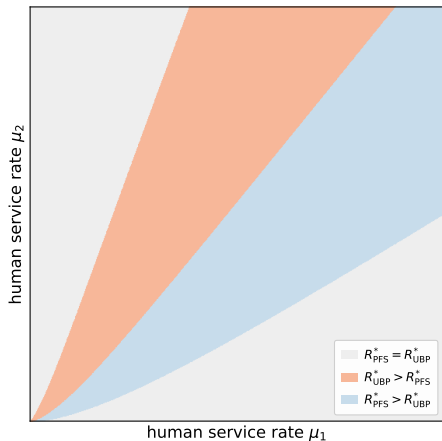


Assumptions: Binary types, $\gamma_2 > \gamma_1$

Main insights.

- 🤖 & 🧑 **Complements:** humans and AI excel at different types; each covers the other's weakness $\Rightarrow$ PFS wins.
- 🤖 & 🧑 **Substitutes:** humans and AI excel at the same types; AI's edge is redundant $\Rightarrow$ UBP wins.

UBP vs. PFS IN THE QUEUEING APPLICATION



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Intuition.

- Task 2 is easier for AI: $\gamma_2 > \gamma_1$.
- lower μ_2 means task 2 becomes more congested if without AI, and thus more valuable to the buyer for using AI
- PFS wins when μ_2 is small: easy-AI task but high buyer valuations

FULL OUTCOME-CONTINGENT PRICING

THEOREM (REPORTING REDUNDANCY)

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This is NOT the revelation principle.

- Revelation principle: without loss, ask the buyer to report the hidden type t rather than other messages.

WHY REPORTING MANIPULATION IS REDUNDANT

THEOREM (REPORTING REDUNDANCY)

With unrestricted outcome-contingent prices, truthful outcome revelation is without loss.

Interpretation

$$\mathbb{E}[\text{bill}] = p_t \cdot \Pr\{\text{"success"}\} + q_t \cdot (1 - \Pr\{\text{"success"}\}) = q_t + (p_t - q_t) \cdot \Pr\{\text{"success"}\}.$$

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**A baseline fee plus success premium
makes truthful reporting credible!**

CONTRACT-DESIGN PROBLEM UNDER TRUTHFULLY REVEALING

$$\begin{aligned} & \max_{\{(a_t, p_t, q_t)\}_{t \in \mathcal{T}}} \sum_t \pi_t \left[(1 - \gamma_t(a_t)) q_t + \gamma_t(a_t) p_t - c(a_t) \right] \\ & \text{s.t.} \quad [\text{IC}] \quad v_t(a_t) - \left[(1 - \gamma_t(a_t)) q_t + \gamma_t(a_t) p_t \right] \\ & \qquad \qquad \geq v_t(a_r) - \left[(1 - \gamma_t(a_r)) q_r + \gamma_t(a_r) p_r \right], \quad \forall t, r \in \mathcal{T}, \\ & \quad [\text{IR}] \quad v_t(a_t) - \left[(1 - \gamma_t(a_t)) q_t + \gamma_t(a_t) p_t \right] \geq 0, \quad \forall t \in \mathcal{T}. \end{aligned}$$

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- If $p_t = q_t$ for all t : standard one-dimensional screening (Mussa and Rosen, 1978; Maskin and Riley, 1984).

REVENUE-MAXIMIZATION MECHANISM

THEOREM (REVENUE-MAXIMIZATION MECHANISM)

- **Reporting mechanism:** truthful outcome revelation is without loss: $R_{\text{SB}} = R_{\text{SB}}^{\text{tr}}$.
- **Contract:** assume $\gamma_t(a) = \alpha_t s(a)$, label types so that $\alpha_1 < \alpha_2 < \dots < \alpha_n$:
 - **endpoint types get first-best allocations:**
 $a_1^{\text{SB}} \in \arg \max_{a \in \mathcal{A}} \{v_1(a) - c(a)\}$ and $a_n^{\text{SB}} \in \arg \max_{a \in \mathcal{A}} \{v_n(a) - c(a)\}$;

WHY ENDPOINT TYPES GET FIRST BEST

Under $\gamma_t(a) = \alpha_t s(a)$, label types so that $\alpha_1 < \alpha_2 < \dots < \alpha_n$.

Lowest endpoint. If type r chooses menu 1, its payoff is

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Thus, endpoint mimics can be deterred by **payments only**:

$$a_t^{\text{SB}} \in \arg \max_{a \in \mathcal{A}} \{v_t(a) - c(a)\}, \quad t \in \{1, n\}.$$

FIRST-BEST EXTRACTION

PROPOSITION (FIRST-BEST EXTRACTION)

Under separability $v_t(a) = \theta_t a$, $\gamma_t(a) = \alpha_t s(a)$, first-best extraction is achieved if ***either***:

- there are only two types;
- the value shifter θ is discretely concave in the success shifter α .

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Intuition.

- **Two types:** both types are endpoints.
- **Concavity:** value heterogeneity is weaker relative to success-probability heterogeneity.
 - the post-allocation outcome is informative enough under concavity

THEOREM (REVENUE-MAXIMIZATION MECHANISM)

- **Reporting mechanism:** truthful outcome revelation is without loss: $R_{SB} = R_{SB}^{tr}$.
- **Contract:** assume $\gamma_t(a) = \alpha_t s(a)$, where $\alpha_1 < \alpha_2 < \dots < \alpha_n$:

– **Endpoint types get first-best allocations:**

$$a_t^{SB} \in \arg \max_{a \in \mathcal{A}} \{v_t(a) - c(a)\}, \quad t \in \{1, n\}.$$

– Further assume $v_t(a) = \theta_t a$: interior types have **curvature virtual values**; for $t = 2, \dots, n-1$,

$$\hat{\theta}_t = \theta_t - \pi_t^{-1} \sum_{r < t < \tau} \mu_t^{r\tau} \Delta_t^{r\tau}, \quad a_t^{SB} \in \arg \max_{a \in \mathcal{A}} \{\hat{\theta}_t a - c(a)\}.$$

- $\mu_t^{r\tau} \geq 0$: IC multiplier for pair $r < t < \tau$ mimicking report t .
- $\Delta_t^{r\tau}$: curvature at t relative to the (r, τ) chord in the (α, θ) profile.

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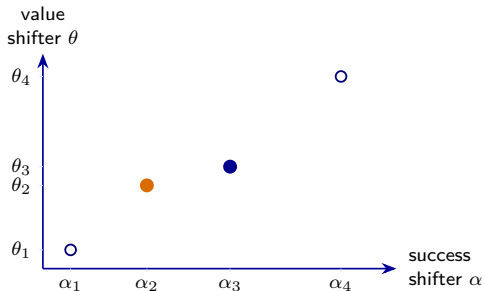
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- $\mu_t^{r\tau} \geq 0$: IC multiplier for pair $r < t < \tau$ mimicking report t .
- $\Delta_t^{r\tau}$: curvature at t relative to the (r, τ) chord in the (α, θ) profile.
- Both directions can happen for interior types:

$$a_t^{SB} < a_t^{FB} \quad \text{or} \quad a_t^{SB} > a_t^{FB}$$

CURVATURE VIRTUAL TYPES

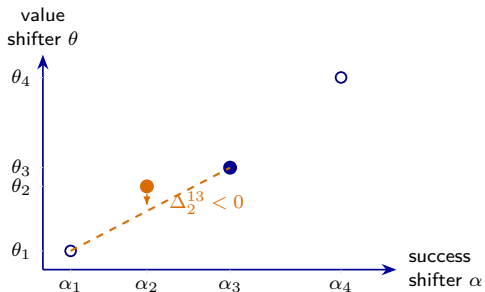
$$\hat{\theta}_t = \theta_t - \pi_t^{-1} \sum_{r < t < \tau} \mu_t^{r\tau} \Delta_t^{r\tau}$$



Here, assume higher success shifter α is associated with higher value shifter θ .

CURVATURE VIRTUAL TYPES

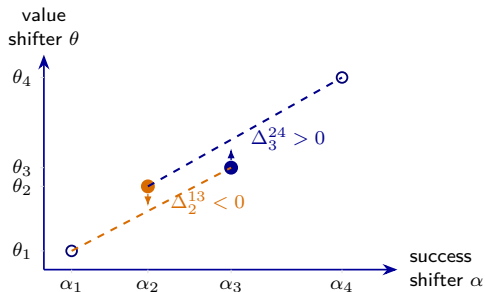
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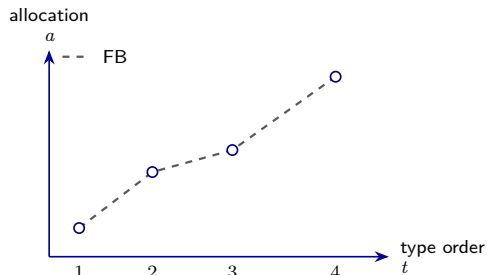
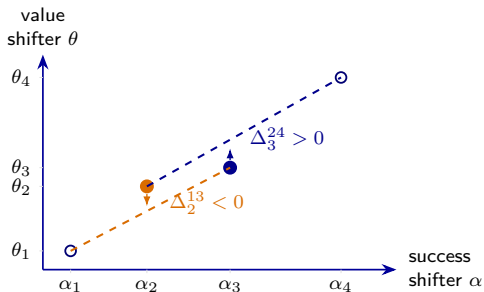
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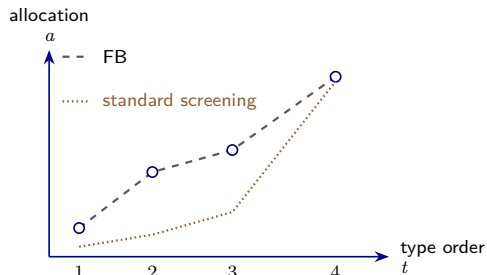
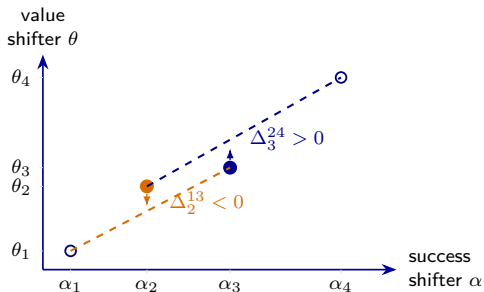
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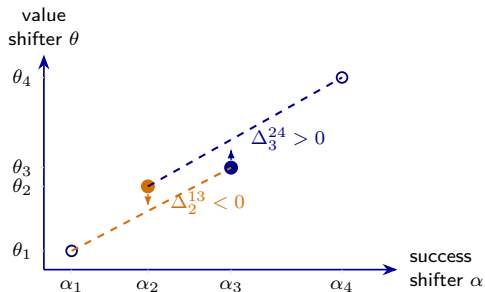
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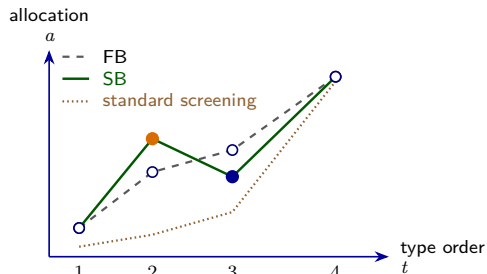
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$$a_t^{\text{SB}} < a_t^{\text{FB}} \quad \text{or} \quad a_t^{\text{SB}} > a_t^{\text{FB}}$$

PROOF SKETCH

ENDPOINT FIRST-BEST ALLOCATIONS

Assume $\gamma_t(a) = \alpha_t s(a)$, and label types so that

$$\alpha_1 < \alpha_2 < \cdots < \alpha_n.$$

Then endpoint types get first-best allocations:

$$a_t^{\text{SB}} \in \arg \max_{a \in \mathcal{A}} \{v_t(a) - c(a)\}, \quad t \in \{1, n\}.$$

ALLOCATION-RENT PROBLEM

Rent identity.

$$u_t = v_t(a_t) - [(1 - \gamma_t(a_t))q_t + \gamma_t(a_t)p_t], \quad \text{expected payment} = v_t(a_t) - u_t.$$

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Seller problem in allocations and rents.

$$\begin{aligned} \max_{\{(a_t, u_t)\}_{t \in \mathcal{T}}} \quad & \sum_{t \in \mathcal{T}} \pi_t [v_t(a_t) - u_t - c(a_t)] \\ \text{s.t.} \quad & [\text{IR}] \quad u_t \geq 0, \quad \forall t \in \mathcal{T}, \\ & [\text{IC}] \quad (a, u) \text{ is implementable.} \end{aligned}$$

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What (a, u) are implementable?

IMPLEMENTABILITY

LEMMA (IMPLEMENTABILITY)

A pair $(a, u) = (a_t, u_t)_{t \in \mathcal{T}}$ is implementable if and only if, for every $t \in \mathcal{T}$, $(a_t, (u_r)_{r \in \mathcal{T}})$ is **upper-supported**.

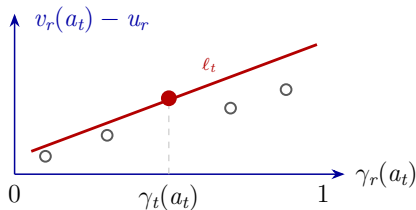
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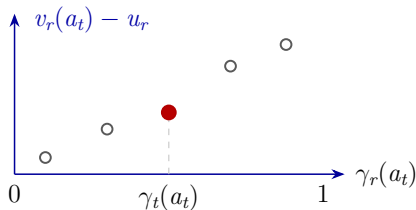
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- $(a_t, (u_r)_{r \in \mathcal{T}})$ is **upper-supported**: $\exists$ a line ℓ_t
- crossing $(\gamma_t(a_t), v_t(a_t) - u_t)$,
 - lying weakly above $(\gamma_r(a_t), v_r(a_t) - u_r), \forall r \neq t$.

Upper-supported



Not upper-supported

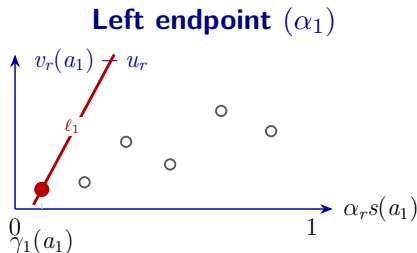


PROOF SKETCH: ENDPPOINT FIRST BEST

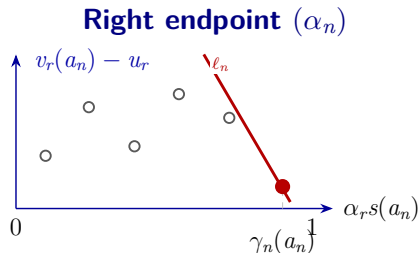
Under $\gamma_t(a) = \alpha_t s(a)$, label types so that $\alpha_1 < \alpha_2 < \dots < \alpha_n$.

- Endpoint reports can be protected by tilting the payment line.
- Thus endpoint allocations carry no IC shadow cost:

$$a_t^{\text{SB}} \in \arg \max_{a \in \mathcal{A}} \{v_t(a) - c(a)\}, \quad t \in \{1, n\}.$$



positive slope deters higher- α mimics



negative slope deters lower- α mimics

SUMMARY

- **AI service pricing creates an outcome-reporting layer.**
 - The seller can fine-tune the AI to misreport failures as billable successes.
- **Simple formats: UBP vs. PFS.**
 - UBP: reporting is irrelevant; this is standard screening.
 - PFS: reporting indeed becomes strategic.
 - PFS is better when buyer-side dispute costs are high, or high-value tasks are AI-easy.
- **Full outcome-contingent pricing makes truthful outcome reporting credible.**
 - Baseline fee plus success premium makes the seller no need to misreport outcomes.
- **Optimal contract.**
 - Success-probability endpoints get first-best allocations.
 - Interior types follow curvature virtual values.

Backup Slides

LARGE κ_B : PFS+ OUTCOME REPORTS $\iff$ LL TWO-PRICE BILLS

PFS + outcome reports

$$\begin{aligned} B_L(\gamma) &= P \cdot \Pr\{\hat{\omega} = \text{success}\} \\ &= P[\gamma\sigma^1 + (1 - \gamma)\sigma^0]. \end{aligned}$$

Limited-liability two-price bill

$$B_R(\gamma) = (1 - \gamma)q + \gamma p, \quad p, q \geq 0.$$

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$$B_L(\gamma) = \gamma p + (1-\gamma)q = B_R(\gamma).$$

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Why limited liability? $p, q \geq 0$ makes $\sigma^1, \sigma^0 \in [0, 1]$.

PROOF SKETCH: UPPER SUPPORT

If type r mimics menu t ,

$$u_r \geq v_r(a_t) - \left[\gamma_r(a_t)p_t + (1 - \gamma_r(a_t))q_t \right].$$

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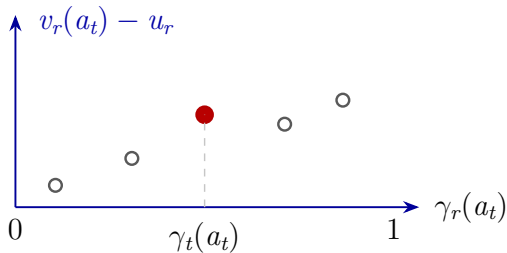
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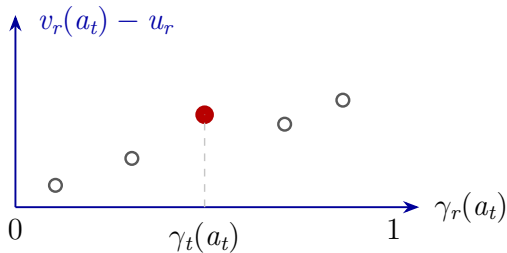


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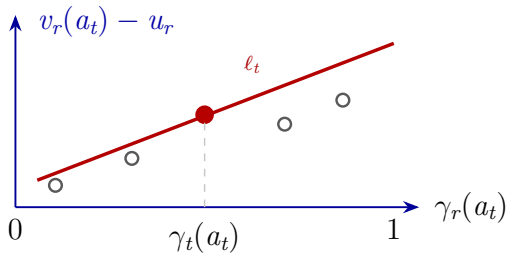
$$\text{expected payment} = q_t + (p_t - q_t)\gamma_r(a_t) = \ell_t(\gamma_r(a_t)).$$

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So IC for menu t is exactly: a line through the true point lies weakly above all mimicker points.

□

PROOF SKETCH

**Why truthful outcome revealing is without loss
under full outcome-contingent prices**

PROOF SKETCH: CONCAVE BUYER COST

- Consider a buyer chooses $(a_t, p_t, q_t, \sigma_t^1, \sigma_t^0)$.
- Given a dispute rule $\delta(\hat{\omega}) \in \{\text{pay}, \text{dispute}\}$,

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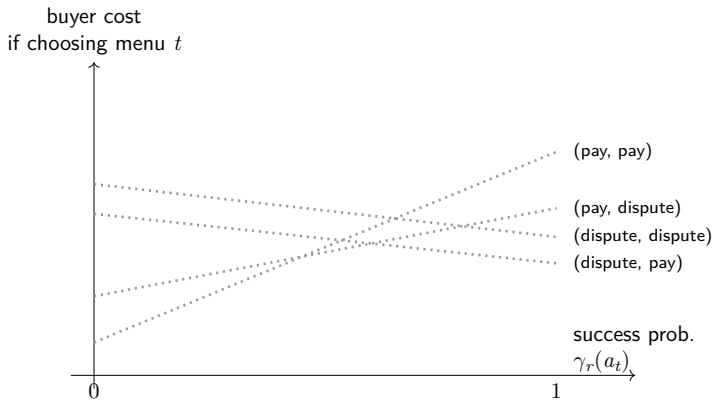
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$$C_{t,\delta}(\gamma) = \gamma C_{t,\delta}^{\text{success}} + (1 - \gamma) C_{t,\delta}^{\text{failure}}$$

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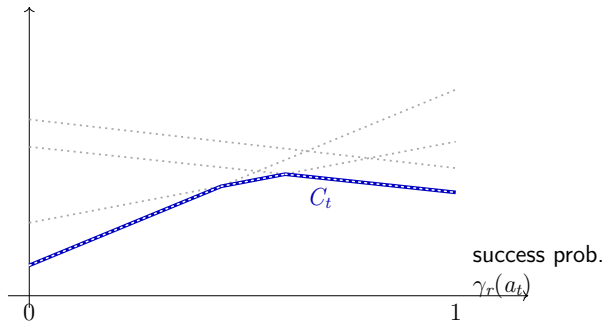
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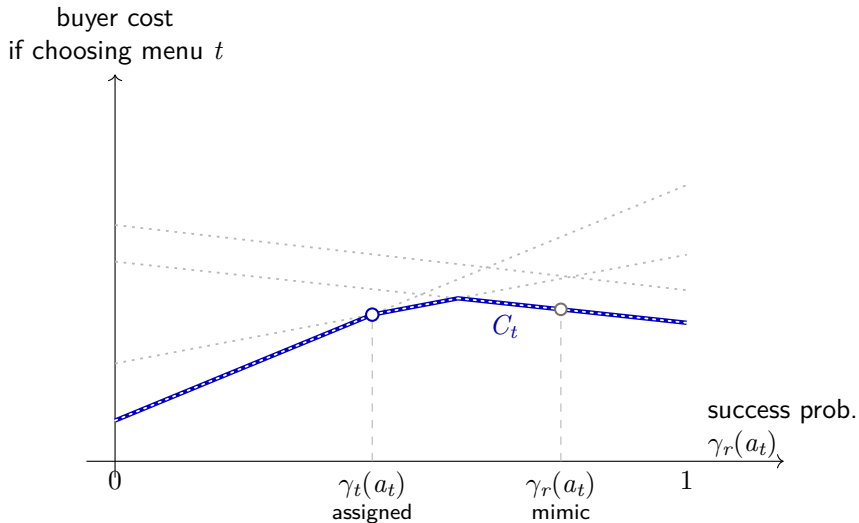
PROOF SKETCH: CONCAVE BUYER COST

buyer cost
if choosing menu t

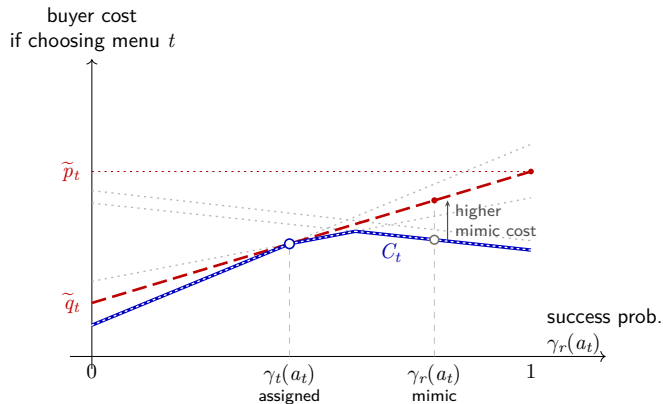


$$C_t(\gamma) = \min_{\delta \in \{\text{pay}, \text{dispute}\}^2} C_{t,\delta}(\gamma)$$

PROOF SKETCH: CONCAVE BUYER COST

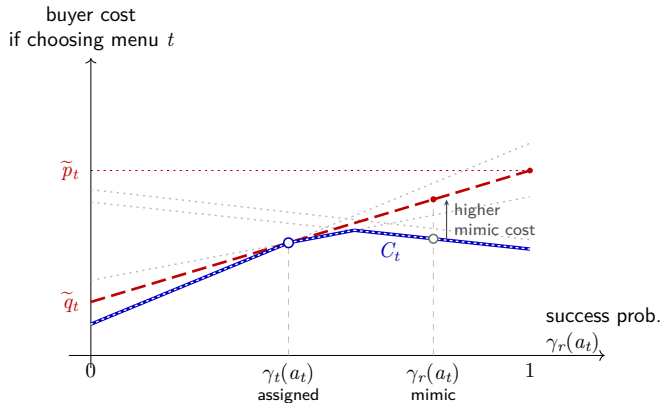


PROOF SKETCH: CONCAVE BUYER COST



- Replace $(a_t, p_t, q_t, \sigma_t^1, \sigma_t^0) \rightsquigarrow (a_t, \tilde{p}_t, \tilde{q}_t, 1, 0)$.
- With truthful revealing, the buyer does not dispute; expected cost equals $\tilde{q}_t + (\tilde{p}_t - \tilde{q}_t)\gamma$.

PROOF SKETCH: CONCAVE BUYER COST



- Buyer t 's expected cost is unchanged.
- Buyer r 's mimic cost becomes higher.
- The seller saves dispute costs.

- Replace $(a_t, p_t, q_t, \sigma_t^1, \sigma_t^0) \rightsquigarrow (a_t, \tilde{p}_t, \tilde{q}_t, 1, 0)$.
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